



Topic 7: Multiple Access

CSE 4405: Data and Telecommunications

Md. Tanvir Hossain Saikat, Jr. Lecturer

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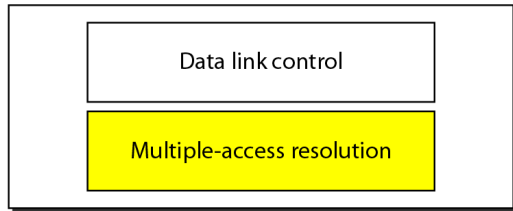
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Data Link Layer and Shared-Media Access

- Data link functionality is often viewed as two sublayers.
- **Upper sublayer:** data-link control tasks.
- **Lower sublayer:** resolves who can use a shared medium.
- Multiple-access protocols belong to this lower access-control role.

Figure: Separates control functions from shared-medium access arbitration.

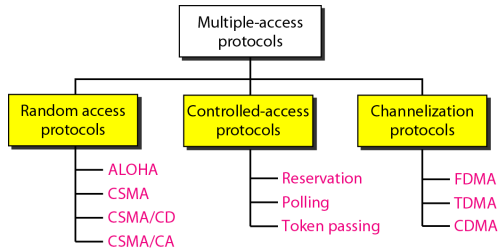
Data link layer



Taxonomy of Multiple-Access Protocols

- When many stations use one broadcast or multipoint link, coordination is required.
- A multiple-access protocol determines how stations share transmission opportunities.
- This chapter organizes methods into random access, controlled access, and channelization.

Figure: Presents the chapter roadmap of access-control families.



12-1 Random Access: Scope

- In contention-based access, no station is permanently superior to others.
- A station with data decides independently whether to transmit, following protocol rules.
- Collisions can occur; each method defines how to reduce and recover from them.

Topics in this section

- ALOHA
- Carrier Sense Multiple Access (CSMA)
- CSMA with Collision Detection (CSMA/CD)
- CSMA with Collision Avoidance (CSMA/CA)

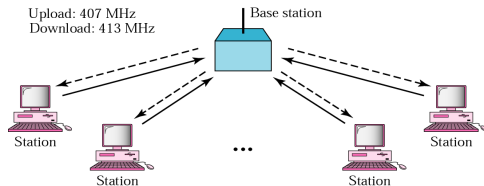
Random Access: Core Questions

- Every station has the right to attempt transmission.
- If two or more stations transmit together, frames collide and become unusable.
- To operate correctly, each station follows a decision procedure:
 - When may I transmit?
 - What if the channel is busy?
 - How do I know success or failure?
 - What do I do after collision?

ALOHA Network: Earliest Random Access

- Developed at the University of Hawaii in the early 1970s.
- Uses a central base station as the communication hop.
- Multiple users share the same incoming channel, so collisions are possible.
- It established foundational ideas for later random-access protocols.

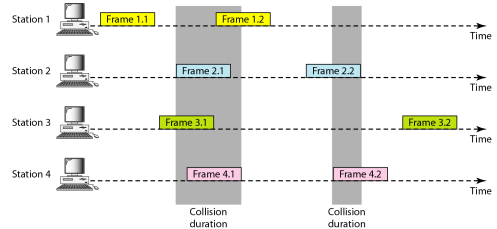
Figure: An early wireless shared-channel architecture where simultaneous attempts can conflict.



Pure ALOHA

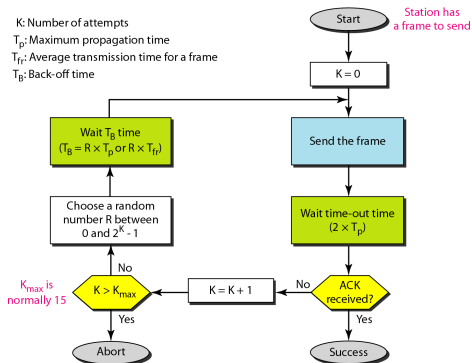
- A station sends whenever it has a frame.
- One common channel is shared by all stations.
- If two frames overlap in time, collision occurs.
- Reliability comes from retransmission after failure.

Figure: Illustrates unslotted transmissions and possible frame overlap.



Procedure for Pure ALOHA

Figure: After transmission, the station waits for feedback; on failure, it backs off randomly and retries.



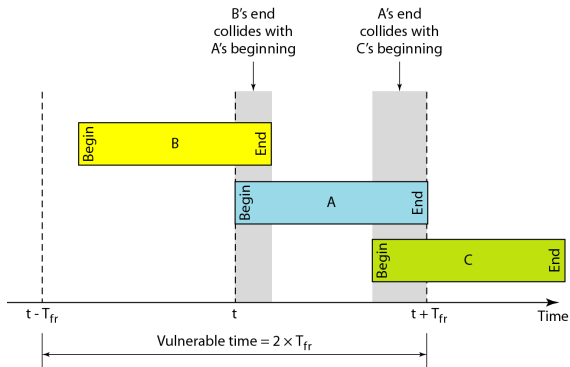
ALOHA Protocol Rule: ACK and Backoff

- After sending, the station waits for an acknowledgment.
- If no ACK arrives within roughly $2T_p$, it assumes loss due to collision.
- Retransmission is delayed by a random backoff T_B , commonly exponential.
- Typical form: $T_B = R \times T_p$ or $T_B = R \times T_{fr}$.
- In binary exponential backoff, $R \in [0, 2^k - 1]$ after k failed attempts.

T_p : maximum propagation time T_{fr} : frame transmission time

Vulnerable Time in Pure ALOHA

Figure: Any overlapping transmission inside a $2T_{fr}$ window can destroy the frame.



Pure ALOHA Throughput

Throughput model: $S = Ge^{-2G}$

Maximum throughput: $S_{\max} = 0.184$ when $G = \frac{1}{2}$

- S : average number of successful transmissions per frame time.
- G : average number of transmission attempts generated per frame time.
- Peak efficiency is about 18.4%, so Pure ALOHA is simple but bandwidth-inefficient.

Pure ALOHA: Probability and Maximum Throughput Derivation

- Let G be the average transmission attempts per frame time T_{fr} , modeled as Poisson.
- Vulnerable interval for one frame is $2T_{fr}$, so the number of competing attempts is

$$X \sim \text{Poisson}(2G).$$

- Success means no competing attempts in that vulnerable interval:

$$P_{\text{succ}} = \Pr[X = 0] = e^{-2G}.$$

- Throughput (successful frames per frame time) is

$$S = GP_{\text{succ}} = Ge^{-2G}.$$

- Maximum occurs at

$$\frac{dS}{dG} = e^{-2G}(1 - 2G) = 0 \Rightarrow G^* = \frac{1}{2},$$

$$S_{\text{max}} = S\left(\frac{1}{2}\right) = \frac{1}{2e} \approx 0.184.$$

Pure ALOHA: Finite- n Derivation $\rightarrow$ Poisson Limit

- Suppose n stations are backlogged; each transmits with probability p in one frame time.
- In Pure ALOHA, vulnerable period is $2T_{fr}$, so all other $n - 1$ stations must stay silent in **two** frame-time windows:

$$P_{\text{succ}|\text{tagged}} = (1 - p)^{2(n-1)}.$$

- Expected successful transmissions per frame time:

$$S_n = np(1 - p)^{2(n-1)}.$$

- Optimize for fixed n :

$$\begin{aligned}\ln S_n &= \ln n + \ln p + 2(n-1) \ln(1-p), \\ \frac{d}{dp} \ln S_n &= \frac{1}{p} - \frac{2(n-1)}{1-p} = 0 \Rightarrow p^* = \frac{1}{2n-1}.\end{aligned}$$

- Therefore

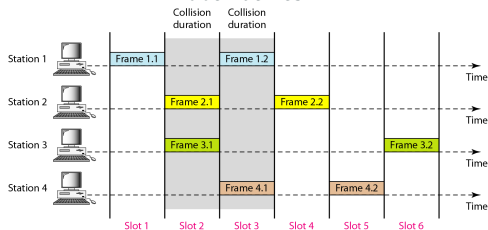
$$S_{n,\max} = \frac{n}{2n-1} \left(\frac{2n-2}{2n-1} \right)^{2n-2}, \quad \lim_{n \rightarrow \infty} S_{n,\max} = \frac{1}{2e}.$$

- With $G = np$ and $n \rightarrow \infty$: $(1 - p)^{2(n-1)} \rightarrow e^{-2G}$, recovering $S = Ge^{-2G}$.

Slotted ALOHA

- Time is divided into slots of duration T_{fr} .
- Stations may transmit only at slot boundaries.
- This prevents partial overlaps and reduces collision window.
- Synchronization is required among participating stations.

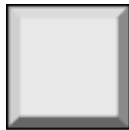
Figure: Shows frame starts aligned to slot boundaries.



Slotted ALOHA Throughput

Throughput model: $S = Ge^{-G}$

Maximum throughput: $S_{\max} = 0.368$ when $G = 1$



- Slotting roughly doubles best-case throughput compared with Pure ALOHA.
- Peak efficiency reaches about 36.8%.

Slotted ALOHA: Probability and Maximum Throughput Derivation

- In each slot (T_{fr}), attempts are Poisson with mean G : $X \sim \text{Poisson}(G)$.
- A slot is successful only when exactly one station attempts:

$$S = \Pr[X = 1] = Ge^{-G}.$$

- Equivalent finite- n model:

$$S = np(1 - p)^{n-1}, \quad G = np.$$

- For fixed n , maximize S by

$$\frac{d}{dp} [np(1 - p)^{n-1}] = 0 \Rightarrow p^* = \frac{1}{n}.$$

- Hence

$$S_{\max,n} = \left(1 - \frac{1}{n}\right)^{n-1}, \quad \lim_{n \rightarrow \infty} S_{\max,n} = \frac{1}{e} = 0.368.$$

- In G -form:

$$\frac{d}{dG} (Ge^{-G}) = e^{-G}(1 - G) = 0 \Rightarrow G^* = 1, \quad S_{\max} = 1/e.$$

Slotted ALOHA: Detailed Probability Steps

- Finite-user model for one slot:

$$\Pr[\text{exactly } k \text{ attempts}] = \binom{n}{k} p^k (1-p)^{n-k}.$$

- A successful slot requires $k = 1$:

$$S_n = \Pr[k = 1] = \binom{n}{1} p (1-p)^{n-1} = np(1-p)^{n-1}.$$

- First derivative:

$$\frac{dS_n}{dp} = n(1-p)^{n-1} - n(n-1)p(1-p)^{n-2} = n(1-p)^{n-2}(1-np).$$

- Thus $p^* = 1/n$, and

$$S_{n,\max} = n \left(\frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} = \left(1 - \frac{1}{n} \right)^{n-1}.$$

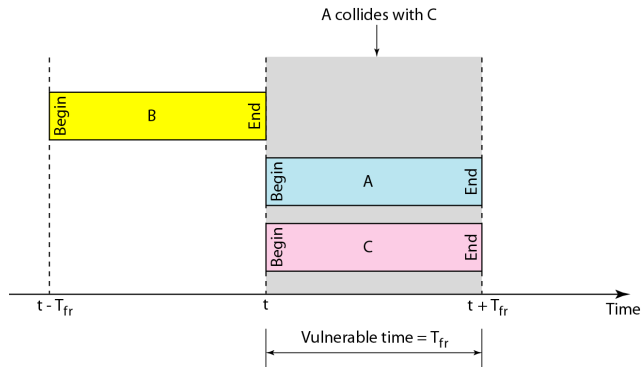
- Poisson limit ($n \rightarrow \infty$, $np = G$):

$$S_n = np(1-p)^{n-1} \rightarrow Ge^{-G},$$

and at $G = 1$, $S_{\max} = 1/e \approx 0.368$.

Vulnerable Time in Slotted ALOHA

Figure: The vulnerable period becomes T_{fr} , which improves success probability.

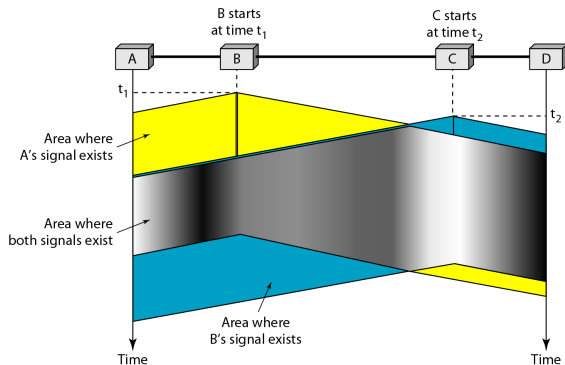


Carrier Sense Multiple Access (CSMA)

- CSMA follows “sense before transmit” (listen before talk).
- By checking channel state first, many avoidable collisions are prevented.
- Collisions still cannot be fully eliminated.
- Reason: propagation delay may hide in-flight transmissions from distant stations.

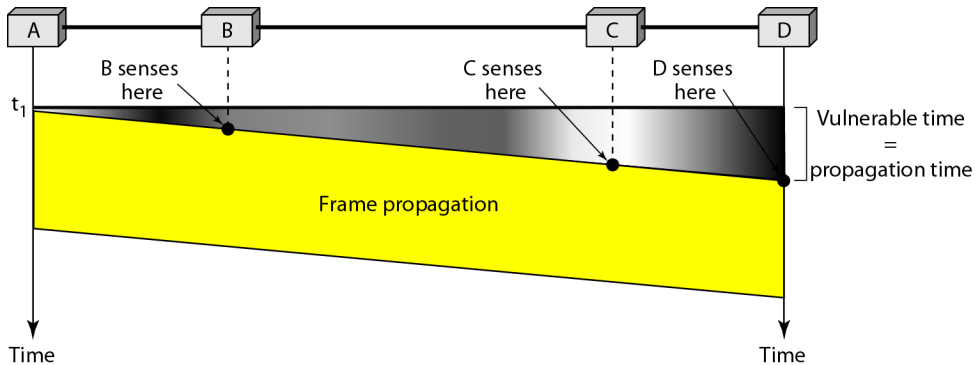
CSMA Collision Space/Time Model

Figure: Two stations can both sense idle and start before each hears the other's first bit.



Vulnerable Time in CSMA

Figure: Propagation delay T_p determines the residual collision window in carrier-sensing systems.



CSMA Throughput Model (Probability-Based Approximation)

- Let n backlogged stations contend after idle sensing; each transmits with probability p .
- Per contention instant:

$$P_i = (1 - p)^n, \quad P_s = np(1 - p)^{n-1}, \quad P_c = 1 - P_i - P_s.$$

- Contention success rate is $S_0 = P_s$. Its maximum is

$$\frac{dP_s}{dp} = n(1 - p)^{n-2}(1 - np) = 0 \Rightarrow p^* = \frac{1}{n},$$

$$S_{0,\max} = \left(1 - \frac{1}{n}\right)^{n-1} \approx \frac{1}{e} \quad (\text{large } n).$$

- Including propagation-delay overhead $a = \frac{T_p}{T_{fr}}$, a common approximation is

$$S_{\text{CSMA}} \approx \frac{S_0}{1 + 2a}, \quad S_{\max} \approx \frac{1/e}{1 + 2a}.$$

- So CSMA improves over ALOHA mainly by shrinking the vulnerable period to a propagation-limited term.

CSMA: Cycle-Time Throughput Derivation (Detailed)

- Write throughput as useful payload time over average contention cycle time:

$$S_{\text{CSMA}} = \frac{P_s T_{\text{payload}}}{P_i \sigma + P_s T_s + P_c T_c}.$$

- Meanings:

$$P_i = (1 - p)^n, \quad P_s = np(1 - p)^{n-1}, \quad P_c = 1 - P_i - P_s.$$

σ : empty sensing slot, T_s : time consumed by one success, T_c : time consumed by one collision.

- Normalize by frame time T_{fr} , set

$$\hat{\sigma} = \frac{\sigma}{T_{fr}}, \quad \hat{T}_s = \frac{T_s}{T_{fr}}, \quad \hat{T}_c = \frac{T_c}{T_{fr}}, \quad \rho = \frac{T_{\text{payload}}}{T_{fr}},$$

then

$$S_{\text{CSMA}} = \frac{\rho P_s}{\hat{\sigma} P_i + \hat{T}_s P_s + \hat{T}_c P_c}.$$

- If $\hat{T}_s \approx \hat{T}_c$ and $\hat{\sigma} \ll \hat{T}_s$, maximizing S_{CSMA} is close to maximizing P_s , giving $p^* \approx 1/n$.
- This yields the practical large- n operating point $G = np \approx 1$, then adjust by delay factor $a = T_p/T_{fr}$.

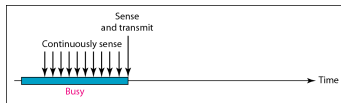
Persistence Methods

- Persistence strategy defines behavior after sensing a busy medium.
- **1-persistent:** transmit immediately once channel is idle.
- **Non-persistent:** wait random time, then sense again.
- **p -persistent:** in slotted operation, transmit with probability p ; defer with $1 - p$.

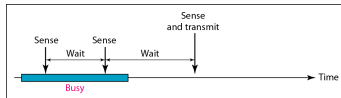
Method	Typical tradeoff
1-persistent	Lower waiting, but higher chance of simultaneous starts
Non-persistent	Fewer collisions, but more idle waiting overhead
p -persistent	Tunable compromise between aggressiveness and caution

Behavior of Persistence Methods

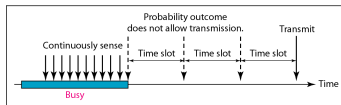
Figure: Different persistence choices reshape delay and collision probability.



a. 1-persistent



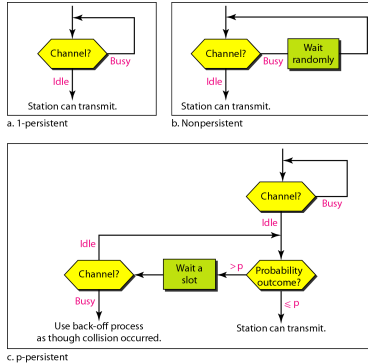
b. Nonpersistent



c. p-persistent

Flow Diagram of Persistence Methods

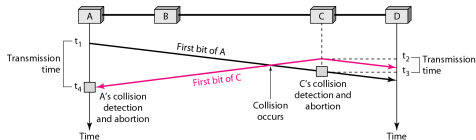
Figure: Carrier sensing plus persistence rules create the station's transmit/defer decision loop.



CSMA with Collision Detection (CSMA/CD)

- Adds collision-handling behavior to CSMA.
- A station monitors the medium while transmitting.
- If collision is detected, it aborts and retries later.
- This was central to classic shared-medium Ethernet.

Figure: Shows detection of first-bit collision in shared-medium access.



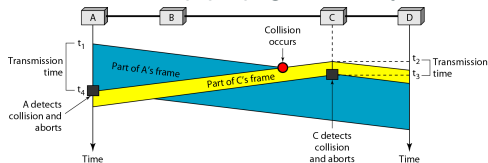
CSMA/CD: Minimum Frame Size Constraint

- For collision detection to work, transmission must last long enough.
- Required condition:

$$T_{fr} \geq 2T_p$$

- Otherwise, a sender might finish before collision evidence returns.

Figure: Relates collision detection to round-trip propagation delay.



CSMA/CD Throughput and Maximum Throughput (Approximate)

- Using the same contention probabilities P_i, P_s, P_c , collision cost is reduced because detection occurs in about $2T_p$, not a full frame.
- Normalize time by T_{fr} , define $a = \frac{T_p}{T_{fr}}$, and mini-slot ratio $\delta = \frac{\sigma}{T_{fr}}$:

$$S_{CD} \approx \frac{P_s}{P_s + 2a P_c + \delta P_i}.$$

- Under saturation ($P_i \approx 0$):

$$S_{CD} \approx \frac{P_s}{P_s + 2a P_c}.$$

- At the contention optimum $p^* = 1/n$ and large n :

$$P_s \approx \frac{1}{e}, \quad P_c \approx 1 - \frac{2}{e}.$$

- Hence an analytical maximum approximation is

$$S_{\max, CD} \approx \frac{1/e}{1/e + 2a(1 - 2/e)} = \frac{1}{1 + 2a(e - 2)}.$$

CSMA/CD: Large- n Optimization View

- Let $x = np$ (attempt intensity per contention instant). For large n :

$$P_i \approx e^{-x}, \quad P_s \approx xe^{-x}, \quad P_c \approx 1 - e^{-x}(1 + x).$$

- Under saturation and negligible idle mini-slot cost:

$$S_{CD}(x) \approx \frac{xe^{-x}}{x} e^{-x} + 2a[1 - e^{-x}(1 + x)].$$

- When $a \rightarrow 0$, denominator $\rightarrow xe^{-x}$, so the maximizing point is the maximizer of xe^{-x} :

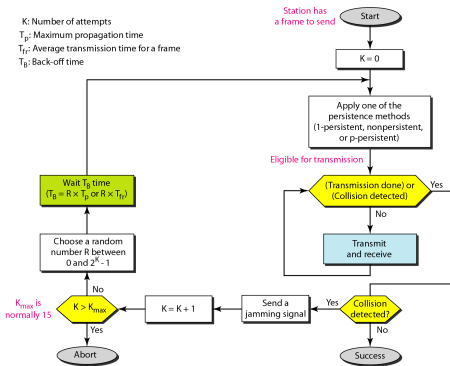
$$\frac{d}{dx}(xe^{-x}) = e^{-x}(1 - x) = 0 \Rightarrow x^* = 1.$$

- For practical $a > 0$, optimum remains near $x \approx 1$ (slightly below/above depending on a).
- Substituting $x = 1$ gives the closed-form estimate

$$S_{CD}(1) \approx \frac{1/e}{1/e + 2a(1 - 2/e)} = \frac{1}{1 + 2a(e - 2)}.$$

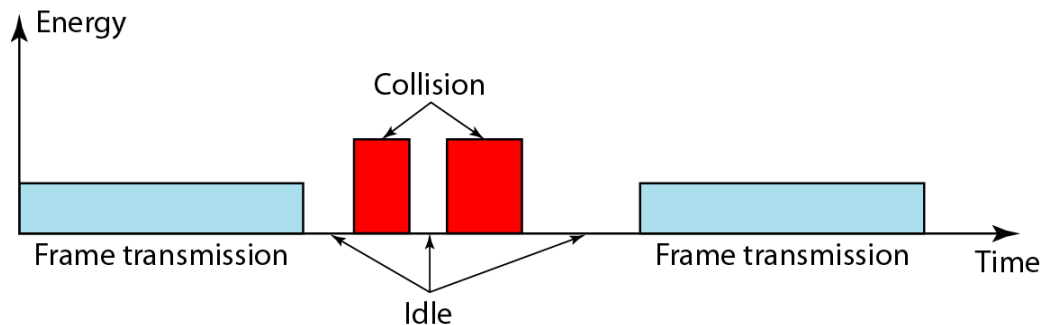
CSMA/CD Flow Diagram

Figure: Sense, transmit, detect, abort, back off, and retry form the CSMA/CD control cycle.



Energy Levels in CSMA/CD

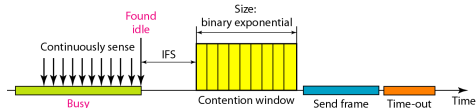
Figure: Collision produces a distinguishable energy pattern used for detection.



CSMA with Collision Avoidance (CSMA/CA)

- Designed for wireless links where collision detection is difficult.
- Avoids collisions using three mechanisms:
 - Interframe Space (IFS)
 - Contention Window (random backoff)
 - Acknowledgments (ACKs)
- The goal is prevention before data overlap occurs.

Figure: Illustrates CSMA/CA timing with IFS and random backoff.



Note

In CSMA/CA, IFS duration can be used to encode priority. Frames/stations with shorter waiting rules may access the channel sooner.



CSMA/CA Note: Busy Channel During Backoff

Note

If the channel becomes busy, a station **pauses** the contention timer; it does not restart from the beginning. Countdown resumes when the channel is idle again.



CSMA/CA Throughput Model and Maximum (DCF-Style)

- Let n stations transmit in a backoff slot with probability τ .

$$P_{tr} = 1 - (1 - \tau)^n, \quad P_{s|tr} = \frac{n\tau(1 - \tau)^{n-1}}{P_{tr}}.$$

- Throughput as payload fraction of average slot duration:

$$S_{CA} = \frac{P_{tr}P_{s|tr}E[P]}{(1 - P_{tr})\sigma + P_{tr}P_{s|tr}T_s + P_{tr}(1 - P_{s|tr})T_c}.$$

- Here T_s includes DATA+ACK+IFS overhead, and T_c includes collision + contention overhead.
- Maximum throughput is obtained by optimizing τ (equivalently CW). Near $\tau^* \approx 1/n$:

$$P_{tr}P_{s|tr} \approx \frac{1}{e}, \quad S_{\max} \approx \frac{(1/e)E[P]}{e^{-1}\sigma + (1/e)T_s + (1 - 2/e)T_c}.$$

- Larger control overhead (T_s, T_c, σ) lowers $S_{\max}$ even if collision probability is controlled.

CSMA/CA: Detailed DCF Probability Chain

- Unconditional success and collision probabilities in one backoff slot:

$$P_s = n\tau(1 - \tau)^{n-1}, \quad P_c = 1 - (1 - \tau)^n - P_s.$$

- Throughput can be written directly with P_s :

$$S_{CA} = \frac{P_s E[P]}{(1 - P_{tr})\sigma + P_s T_s + P_c T_c}, \quad P_{tr} = 1 - (1 - \tau)^n.$$

- If the denominator variation is slow around operating point, first-order maximization is dominated by P_s :

$$\frac{d}{d\tau} [n\tau(1 - \tau)^{n-1}] = n(1 - \tau)^{n-2}(1 - n\tau) = 0 \Rightarrow \tau^* \approx \frac{1}{n}.$$

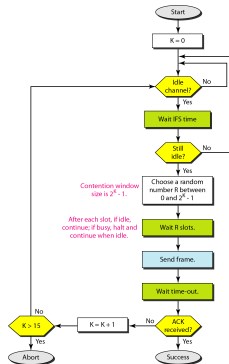
- Practical link to contention window (basic approximation):

$$\tau \approx \frac{2}{CW_{\min} + 1} \Rightarrow CW_{\min}^* \approx 2n - 1.$$

- So tuning CW controls τ , which controls P_s, P_c , and therefore the throughput maximum.

CSMA/CA Flow Diagram

Figure: Carrier sensing, timer freezing/resuming, and ACK-based retry drive wireless contention control.



12-2 Controlled Access: Scope

- Stations consult one another to decide who may transmit.
- A station transmits only when authorized by protocol rules.
- This reduces collisions at the cost of coordination overhead.

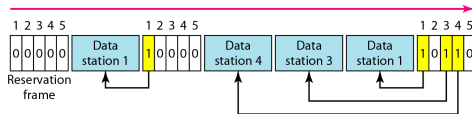
Topics in this section

- Reservation
- Polling
- Token Passing

Reservation Access Method

- Each interval begins with a reservation frame.
- With N stations, reservation frame has N minislots.
- Station i reserves by marking its own minislot.
- Reserved stations transmit data after reservation phase.

Figure: Separates contention-free reservations from the following data phase.

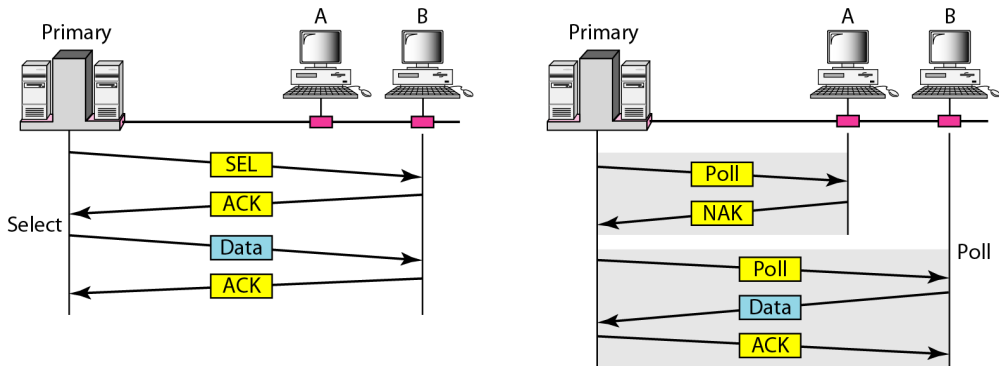


Polling Access Method

- One station is designated **primary**; others are **secondary**.
- The primary controls link operation and sequencing.
- **Poll function:** primary asks secondaries whether they have data to send.
- **Select function:** primary instructs a secondary to prepare to receive.
- Secondary stations follow primary control messages.

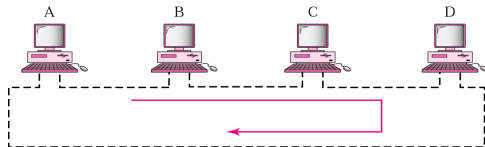
Select and Poll Functions

Figure: Primary station alternates between soliciting uplink data and assigning downlink reception.



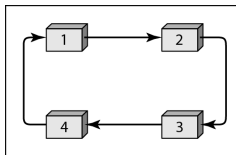
Token-Passing Network

- Transmission right is represented by a special frame: the **token**.
- A station may send data only while holding the token.
- After transmission, token is passed to the next station.
- This enforces orderly, collision-free access.

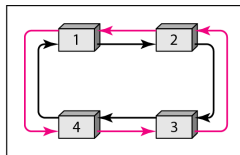


Logical Ring vs Physical Topology

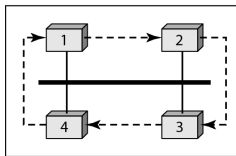
Figure: Token circulation follows a logical order that may differ from physical cabling layout.



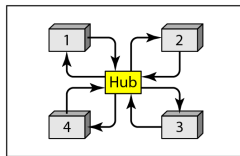
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

12-3 Channelization: Scope

- Channelization shares one link by partitioning resources among stations.
- Partitioning can be by **frequency**, **time**, or **code**.
- Unlike contention methods, structured separation is built in.

Topics in this section

- Frequency-Division Multiple Access (FDMA)
- Time-Division Multiple Access (TDMA)
- Code-Division Multiple Access (CDMA)

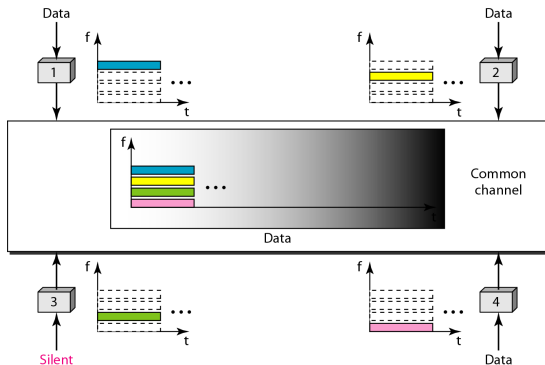
- Total available bandwidth is shared among stations by frequency bands.
- FDMA is a data-link access method that relies on FDM at the physical layer.
- Guard bands separate adjacent user bands to reduce interference.
- Each station keeps its assigned band while active.

Key statement

In FDMA, one common channel is split into frequency bands separated by guard bands.

Frequency-Division Multiple Access

Figure: Users transmit simultaneously in distinct frequency slices of the shared spectrum.



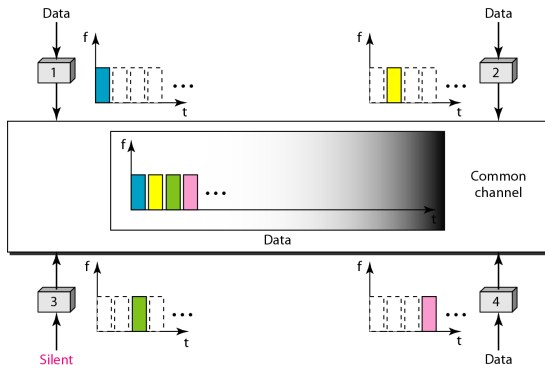
- Entire system bandwidth is treated as one channel.
- Stations share that channel in time.
- Each station transmits in assigned slot(s) within repeating frames.
- Timing alignment is essential for correct slot separation.

Key statement

In TDMA, one channel is time-shared among stations.

Time-Division Multiple Access

Figure: Users reuse the same frequency resource by transmitting in different time slots.



- One channel occupies the entire link bandwidth.
- All stations can transmit simultaneously.
- Separation is achieved by distinct spreading codes, not time or frequency slices.
- CDMA therefore differs fundamentally from both FDMA and TDMA.

Key statement

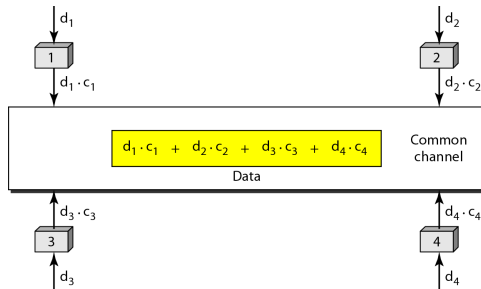
In CDMA, one shared channel carries all transmissions at once using code separation.

Simple Communication Idea in CDMA

- Assigned codes are designed to be orthogonal.
- Cross-correlation of different codes is 0.
- Auto-correlation of a code with itself is N (number of stations/chips).

$$(d_1 c_1 + d_2 c_2 + d_3 c_3 + d_4 c_4) \cdot c_1 = 4d_1$$

Figure: Shows how orthogonality enables one user's recovery from a composite signal.



CDMA Chip Sequences

- CDMA is built on coding theory.
- Each station receives a chip sequence (code).
- Code properties enable simultaneous transmission and separation.

Figure: Provides example chip sequences assigned to stations.



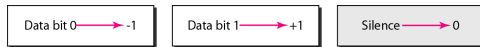
Properties of Orthogonal Sequences

- For identical sequences A : $A \cdot A = N$
- For two different orthogonal sequences A, B : $A \cdot B = 0$
- For sign-inverted sequence: $A \cdot (-A) = -N$
- These identities enable despreading and user separation at the receiver.

CDMA Data Representation

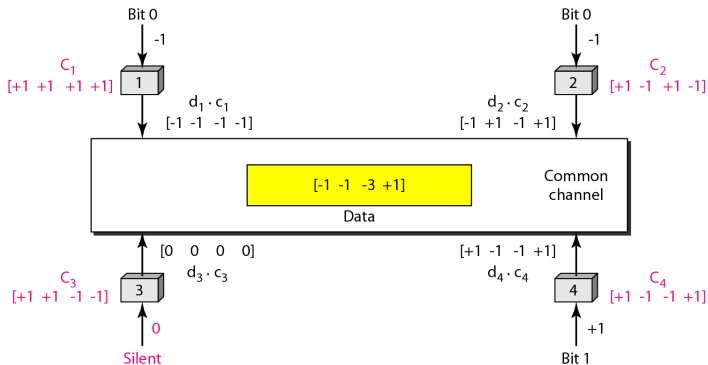
- Data bits are mapped to chip-level signal representations.
- Idle condition is represented by zero signal contribution.
- Active bits are spread according to assigned chip code.

Figure: Links bit values and idle state to CDMA signal representation.



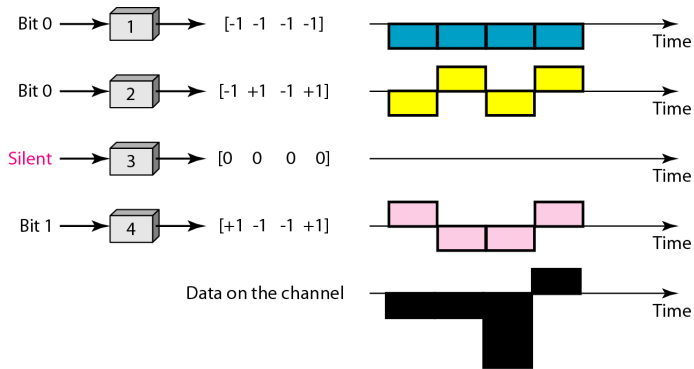
Encoding and Decoding: Sharing a CDMA Channel

Figure: Station signals are encoded, summed, and later decoded using code correlation.



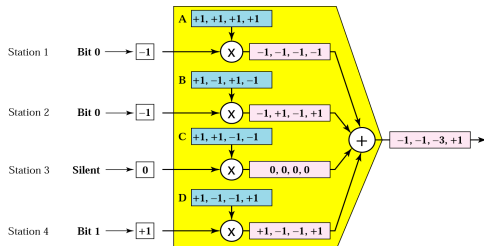
Composite Digital Signal in CDMA

Figure: The transmitted waveform is the superposition of simultaneously spread user signals.



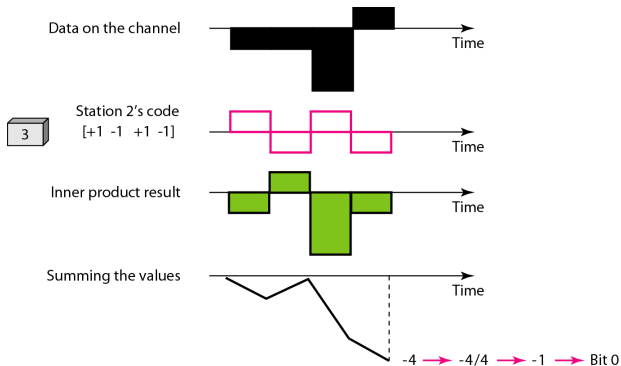
CDMA Encoding Rules: Multiplexer View

- Consider four stations during one bit interval.
- Each station multiplies its bit by its chip sequence.
- The CDMA multiplexer adds all encoded chip streams.
- Result is one composite signal on the shared link.



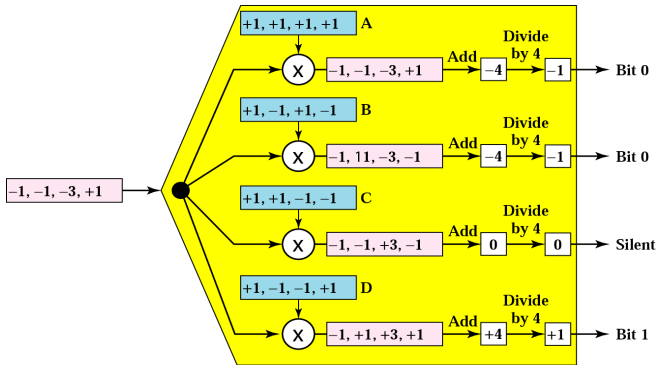
CDMA Decoding Example for One Station

Figure: Correlating the composite signal with one station's code extracts that station's data value.



CDMA Encoding Rules: Demultiplexer View

Figure: The demultiplexer applies user-specific codes to separate parallel data streams from the sum signal.



CDMA Sequence Generation with Walsh Tables

- Walsh tables are square matrices of orthogonal chip sequences.
- Each row can be assigned as a user code.
- Table construction follows recursive orthogonal rules.

Figure: Shows the general Walsh-table construction pattern and examples.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1 , W_2 , and W_4

CDMA Note: Required Number of Sequences

Note

The number of sequences in a Walsh table must satisfy $N = 2^m$, where m is a nonnegative integer.



Example 12.6: Chips for 2 and 4 Stations

Problem Find chip sequences for:

- (a) Two-station network
- (b) Four-station network

Solution using Walsh rows

- W_2 : $[+1, +1]$, $[+1, -1]$
- W_4 : $[+1, +1, +1, +1]$, $[+1, -1, +1, -1]$, $[+1, +1, -1, -1]$, $[+1, -1, -1, +1]$

Example 12.7: Number of Sequences for 90 Stations

Problem How many orthogonal sequences are needed for 90 stations?

Solution

- Walsh size must be power of two: $N = 2^m$.
- Smallest $2^m \geq 90$ is $2^7 = 128$.
- So choose $m = 7$, generate 128 sequences, and assign 90 of them.

Summary (1)

- Data link layer can be viewed as control sublayer plus shared-medium access sublayer.
- Multiple-access protocols are grouped as random access, controlled access, and channelization.
- In random access, stations contend without fixed authority.
- ALOHA enables decentralized access but is collision-prone.
- Simultaneous transmissions can corrupt frames on the shared medium.

Summary (2)

- CSMA improves performance by sensing before transmitting.
- Persistence methods include 1-persistent, non-persistent, and p -persistent behavior.
- CSMA/CD extends CSMA with collision detection and retransmission logic.
- CSMA/CA targets wireless systems using IFS, contention windows, and ACKs to avoid collisions.

Summary (3)

- In controlled access, stations coordinate authorization before transmission.
- Reservation uses minislots to request future data transmission opportunities.
- Polling relies on a primary controller with poll/select operations.
- Token passing circulates a token in a logical ring to grant send permission.

Summary (4)

- Channelization shares resources by frequency, time, or code.
- FDMA allocates fixed frequency bands.
- TDMA allocates recurring time slots.
- CDMA assigns orthogonal chip sequences and allows simultaneous transmissions.
- Walsh tables provide practical orthogonal sequence generation for CDMA.

Questions and Discussion

- Random vs controlled access tradeoffs
- CSMA/CD and CSMA/CA differences
- FDMA, TDMA, and CDMA comparison

